

Lesson 18: Introduction to Partial Differential Equations & Differential Operators

Professor Dave Explains Differential Equations Master Series · Lesson 18

Video: Introduction to Partial Differential Equations: Classification and Differential Operators

Course: Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains that ordinary differential equations describe point particles moving through time (0D spatial points). But the real universe consists of continuous fields: heat diffusing through an engine block, acoustic waves vibrating a concert hall, and gravitational fields warping spacetime. These require Partial Differential Equations (PDEs), which involve rates of change with respect to space and time simultaneously. Just like conic sections in geometry, second-order linear PDEs classify into Elliptic (steady state), Hyperbolic (waves), and Parabolic (diffusion).

2. Core Theoretical Framework & Rigorous Formulations

1. Second-Order Linear PDE Classification

For the general two-variable linear equation $Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = G$:

Discriminant $\Delta = B^2 - 4AC$	Classification	Physical Prototype	Qualitative Behavior
$\Delta < 0$	Elliptic	Laplace's Eq: $u_{xx} + u_{yy} = 0$	Smooth steady-state equilibrium; no real characteristics.
$\Delta > 0$	Hyperbolic	Wave Eq: $u_{tt} - c^2 u_{xx} = 0$	Disturbances propagate along characteristics at finite speed c .
$\Delta = 0$	Parabolic	Heat/Diffusion Eq: $u_t - \alpha^2 u_{xx} = 0$	Dissipative smoothing; infinite speed of propagation.

2. Differential Vector Operators

$$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right), \quad \nabla^2 = \Delta = \nabla \cdot \nabla = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

3. Fully Solved Whiteboard Problems

Concordia PDE Drill: Spatial Classification of Variable-Coefficient PDE

Problem Statement: Classify the PDE $xu_{xx} + u_{yy} = 0$ throughout the 2D Cartesian plane $\mathbb{R}^2$.

Step 1: Identify Discriminant Coefficients

$A = x, \quad B = 0, \quad C = 1.$

Step 2: Compute Discriminant Formula

$\Delta(x, y) = B^2 - 4AC = 0^2 - 4(x)(1) = -4x.$

Step 3: Analyze Regions in the Cartesian Plane

- For $x > 0$: $\Delta = -4x < 0 \implies$ The equation is **Elliptic** (behaves like Laplace's potential equation in the right half-plane).
- For $x < 0$: $\Delta = -4x > 0 \implies$ The equation is **Hyperbolic** (behaves like the wave equation in the left half-plane).
- For $x = 0$ (the y -axis): $\Delta = 0 \implies$ The equation is **Parabolic** (the transition boundary curve, known as the Tricomi equation in transonic aerodynamics!).

⚠ Common Exam Pitfall & Trapdoor Avoidance

The coefficients A, B, C may vary across space. An equation can be elliptic in one region, parabolic on a curve, and hyperbolic elsewhere. Always evaluate the sign of $B^2 - 4AC$ as a function of coordinates.

4. Key Takeaways & Exam Strategy Formula Card

🔑 Core Takeaway & Exam Rule

$B^2 - 4AC < 0$ is Elliptic (Laplace); $B^2 - 4AC > 0$ is Hyperbolic (Wave); $B^2 - 4AC = 0$ is Parabolic (Diffusion).

- Video reference: [Introduction to Partial Differential Equations: Classification and Differential Operators](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.